

A Healthy Mind in a Healthy Body

Lifestyle Medicine Small Group

Department of Medical Humanities & History of Medicine

ER Module

Description

Self-awareness of stress levels and other personal domains relevant to health is important for optimal academic outcomes and for professional and personal development beyond graduation. This professional development session integrates lecture content from the Neurobehavioral Module and utilizes self-assessment and practical tools to aid students in gaining self-awareness and selecting appropriate interventions to improve long-term well-being and academic performance. Outcomes include improved understanding of lifestyle medicine and how to design a holistic, systems-oriented assessment and wellness plan for yourself and patients.

1. Learning Objectives

- SOM.PF.I.BPM2.9.NB.4.BEHS.0169: Apply self-assessment tools in the domains of stress, anxiety, depression, physical activity, and sleep to design a holistic individualized treatment plan
- SOM.PF.I.BPM2.9.NB.4.BEHS.0170: Apply behavioral interventions designed to decrease stress and increase self-health-promoting behaviors
- SOM.PF.I.BPM2.9.NB.4.BEHS.0171: Illustrate why behavioral changes, such as increasing physical activity, can be difficult, using a case study about a patient who has to make health-related changes, and self-reflect on what challenges they may face.

2. Activities

Part 1 (10-minute self-assessment phase). This self-assessment exercise will involve answering questions from commonly used measures of stress, anxiety, depression, physical activity, and sleep. Questionnaires are provided with the small group material and should ideally be completed before attending the session. Please bring your completed forms to the session in either electronic or printed form. After answering all questions, students will calculate their final scores and complete the results page to summarize their personalized lifestyle medicine profile.

These questionnaires serve as a guide to help you see where you are. None of them in isolation is meant to provide a medical diagnosis of any kind and should not be considered as such. While these are validated clinical tools, their purpose is a self-check to create awareness.

Part 2 (45-minute training in Lifestyle Medicine interventions/treatment). This phase involves active participation in physical activity, breathing exercises, progressive muscle relaxation, and behavioural methods for improved sleep hygiene.

Part 3 (10-minute group work). This activity involves group participation in using a case study to discuss barriers to making health-related, behavioral changes.

Part 4 (10-minute self-reflection). This exercise involves self-reflection to assess motivation, readiness to change, support systems, and possible barriers to implementing life-style changes.

PART 1 SELF-ASSESSMENT

SELF-ASSESSMENT TOOLS

1. Depression, Anxiety, Stress Scale (DASS)
2. Physical Activity Scale
3. Pittsburgh Sleep Quality Index (PSQI)

1. Depression, Anxiety, Stress Scale (DASS)

Please read each statement and circle a number 0, 1, 2 or 3 which indicates how much the statement applied to you **over the past week**. There are no right or wrong answers. Do not spend too much time on any statement.

The rating scale is as follows:

0 Did not apply to me at all (0 points)

1 Applied to me to some degree, or some of the time (1 point)

2 Applied to me to a considerable degree or a good part of time (2 points)

3 Applied to me very much or most of the time (3 points)

1	I found it hard to wind down	0	1	2	3
2	I was aware of dryness of my mouth	0	1	2	3
3	I couldn't seem to experience any positive feeling at all	0	1	2	3
4	I experienced breathing difficulty (e.g. excessively rapid breathing, breathlessness in the absence of physical exertion)	0	1	2	3
5	I found it difficult to work up the initiative to do things	0	1	2	3
6	I tended to over-react to situations	0	1	2	3
7	I experienced trembling (e.g. in the hands)	0	1	2	3
8	I felt that I was using a lot of nervous energy	0	1	2	3
9	I was worried about situations in which I might panic and make a fool of myself	0	1	2	3
10	I felt that I had nothing to look forward to	0	1	2	3
11	I found myself getting agitated	0	1	2	3

12	I found it difficult to relax	0	1	2	3
13	I felt down-hearted and blue	0	1	2	3
14	I was intolerant of anything that kept me from getting on with what I was doing	0	1	2	3
15	I felt I was close to panic	0	1	2	3
16	I was unable to become enthusiastic about anything	0	1	2	3
17	I felt I wasn't worth much as a person	0	1	2	3
18	I felt that I was rather touchy	0	1	2	3
19	I was aware of the action of my heart in the absence of physical exertion (e.g. sense of heart rate increase, heart missing a beat)	0	1	2	3
20	I felt scared without any good reason	0	1	2	3
21	I felt that life was meaningless	0	1	2	3

DASS-21 Scoring Instructions

The DASS-21 is comprised of 3 scales (depression, anxiety, and stress), which each contain 7 items. The depression scale assesses dysphoria, hopelessness, devaluation of life, self-deprecation, lack of interest / involvement, anhedonia and inertia. The anxiety scale assesses autonomic arousal, skeletal muscle effects, situational anxiety, and subjective experience of anxious affect. The stress scale is sensitive to levels of chronic non-specific arousal. It assesses difficulty relaxing, nervous arousal, and being easily upset / agitated, irritable / over-reactive and impatient.

Scores for depression, anxiety and stress are calculated by summing the scores for the relevant items. The relevant items for each scale are indicated in the columns below. Please fill in your scores for each of the items listed below to obtain your sum total for each of the three scales.

Depression Scale

Item #3 _____

Item #5 _____

Item #10 _____

Item #13 _____

Item #16 _____

Item #17 _____

Item #21 _____

Depression Sum _____

Anxiety Scale

Item #2 _____

Item #4 _____

Item #7 _____

Item #9 _____

Item #15 _____

Item #19 _____

Item #20 _____

Anxiety Sum _____

Stress Scale

Item #1 _____

Item #6 _____

Item #8 _____

Item #11 _____

Item #12 _____

Item #14 _____

Item #18 _____

Stress Sum _____

2. PHYSICAL ACTIVITY SCALE

How many minutes of regular, specific and structured exercise do you engage in per week?

Explanation:

- **‘Regular’** refers to sporting activities that one undertakes REGULARLY, i.e. scheduled on specific days/time inside a week (e.g. swimming every Tuesday). In contrast, ‘gardening when my lawn needs cutting’ would not be considered regular.
- **‘Specific’** refers to the content of the sporting activities, i.e. does the activity aim to improve specific fitness components? For example, a Pilates class would qualify as it does improve core strength, but a walk in the park when the weather is nice would not qualify.
- **‘Structured’** also refers to the content of the sporting activity. All activities led by a professional, i.e. any (group) exercise led by an exercise instructor or personal trainer, as well as activities undertaken as part of a sports club, would qualify. Going to the gym regularly would also count, as people who are actively paying for/using their subscription normally train at the desired level. However, if participants report that they go to the gym mainly to chat with their peers, the activity would not count towards weekly exercise time.

3. PITTSBURG SLEEP QUALITY INDEX (PSQI)

Instructions: This scale is to assess your usual sleep habits during the **past month ONLY**. Your answers should indicate the most accurate reply for the majority of days and nights in the past month. Please answer all questions.

RESULTS: PERSONALIZED LIFESTYLE MEDICINE PROFILE

Please transfer your total scores for each scale and refer to the clinical scale ranges below:

Depression	_____
Anxiety	_____
Stress	_____
Physical Activity	_____
Sleep (PSQI)	_____

1. DASS Depression Scale: A score of 0-4 is within normal limits; 5-6 suggests a mild degree of depression; 7-10 suggests a moderate degree of depression; 11-13 suggests a high degree of depression; 14+ suggests a severe degree of depression and clinical attention may be warranted.

2. DASS Anxiety Scale: A score of 0-3 is within normal limits; 4-5 suggests a mild degree of anxiety; 6-7 suggests a moderate degree of anxiety; 8-9 suggests a high degree of anxiety; 10+ suggests a severe degree of anxiety and clinical attention may be warranted.

3. DASS Stress Scale: A score of 0-7 is within normal limits; 8-9 suggests a mild degree of stress; 10-12 suggests a moderate degree of anxiety; 13-16 suggests a high degree of stress; 17+ suggests a severe degree of stress and modification of lifestyle factors to reduce stress is recommended.

Note on interpreting DASS-21 index scores: The DASS-21 is based on a dimensional rather than a categorical conception of psychological disorder. The assumption on which the DASS-21 development was based (and which was confirmed by the research data) is that the differences between the depression, anxiety and the stress experienced by normal subjects and clinical populations are essentially differences of degree. The DASS-21 therefore has no direct implications for the allocation of patients to discrete diagnostic categories postulated in classificatory systems such as the DSM and ICD.

4. Physical Activity Scale: The World Health Organization recommends at least 150 minutes of exercise per week. Any number below 150 indicates the need to increase time spent exercising.

5. Pittsburg Sleep Quality Index: Seven component scores are derived, with a score of 0 indicating no difficulty, and a score of 3 indicating the most negative or problematic sleep quality. The component scores are added to produce a global score (range 0 – 21). Higher scores indicate worse sleep quality.

Part 2 Interventions

1. Square Breathing

- Guided group exercise

A good overview can be found here: (<https://www.healthline.com/health/box-breathing>)

2. Mindfulness

- Mindfulness based therapy is a practice that uses Buddhist and Yogic ideas. It is grounded in non-judgemental awareness of self and environment as well as an attitude of open acceptance and curiosity.
- The premise of mindfulness-based therapy is experiencing each moment non-judgmentally and with an open mind. This is done through mindful meditation; yoga stretches and exercises, body scans and breath work. This modality is said to oppose the effect of stressors as it reduces preoccupation with the past or the future which exacerbates depression and anxiety. The deep breathing associated with these practices are also helpful in reducing stress, as it helps to balance the sympathetic and parasympathetic responses.
- Mindfulness-based meditation exercises are used to:
 - i. Decrease physiological arousal
 - ii. Improve moment-to-moment awareness of what is happening in the mind and body
 - iii. Develop non-judgmental acceptance
 - iv. Sensing and feeling without “doing”
 - v. Replace negative emotional states with relaxation

3. Progressive Muscle Relaxation

- Guided group exercise

4. Physical Activity Guidelines (World Health Organization)

- Adults aged 18–64 should do at least 150 minutes of moderate-intensity aerobic physical activity throughout the week or do at least 75 minutes of vigorous-intensity aerobic physical activity throughout the week or an equivalent combination of moderate- and vigorous-intensity activity.
- Aerobic activity should be performed in bouts of at least 10 minutes duration.
- For additional health benefits, adults should increase their moderate-intensity aerobic physical activity to 300 minutes per week, or engage in 150 minutes of vigorous-intensity aerobic physical activity per week, or an equivalent combination of moderate- and vigorous-intensity activity.
- Muscle-strengthening activities should be done involving major muscle groups on 2 or more days a week.

5. Sleep Hygiene Recommendations

- Use bed only to sleep (with exception of sex)
 - i. Lie down only when tired
 - ii. If not asleep in 10 min, depart bed and return only when tired
 - iii. If you wake up early, don't stay in bed
- Maintain consistent sleep schedule
- Avoid caffeine, esp. after noon
- Exercise *more than* 2 hrs before bedtime
- Take hot bath *within* 2 hrs of bedtime
- Create an environment conducive to sleep, including:
 - Cool bedroom temperature (via a/c, fans)
 - Bedroom darkness (blackout curtains, eye shades)
 - Bedroom quietness (white noise machines [fans, humidifiers] earplugs)
- Avoid stimuli not conducive to sleep, including avoidance of (or at least limiting):
 - Bright light from cell phones, computer screens, and TV screens (some recommend turning off devices at least 30 minutes before bedtime)
 - Night time alcohol (facilitates sleep onset, but disrupts sleep architecture)
 - Night time fluids (restricting fluid intake can pre-empt night time awakenings for bathroom breaks)
 - Foods causing indigestion (e.g., some people experience indigestion from rich foods, fatty/fried meals, spicy dishes, citrus fruits), which can cause heartburn and disrupt sleep.
- Use relaxation techniques such as diaphragmatic breathing and progressive muscle relaxation

6. Nutrition

- Pay attention to what you eat
- Focus on nutrient-dense foods
- Eat at regular times every day
- Eat enough!!

7. Group Activity

- Using the case study below, please complete the following:
 - i. Discuss and note at least 2 possible barriers to making these behavioral changes. Barriers may include physical, social, physiological and/or financial.
 - ii. Be prepared to share with the class.

Case Study: James – Managing Type 2 Diabetes Through Lifestyle Change

Patient Profile:

- **Name:** James Carter
- **Age:** 52
- **Occupation:** Delivery Driver
- **Medical Condition:** Recently diagnosed with Type 2 Diabetes
- **Primary Goal:** Improve blood sugar control through diet, exercise, and weight loss

Background:

James was diagnosed with Type 2 diabetes during a routine check-up. His doctor recommended immediate lifestyle changes, including adopting a low-carb diet, increasing physical activity, and losing 20 pounds over the next six months.

Recommended Lifestyle Changes:

1. **Dietary Adjustments:** Shift to a low-glycemic, high-fiber diet.
2. **Physical Activity:** At least 30 minutes of moderate exercise five days a week.
3. **Weight Management:** Gradual weight loss through calorie control and activity.
4. **Medication Adherence:** Daily metformin and regular glucose monitoring.

8. Self-Reflection

- ✓ Take a moment to think about a health habit you would like to adopt or change
 - i. Why do you want to make the change?
 - ii. What would your life be like with this change implemented?
- ✓ Think about what motivates you to make this change
 - iii. How motivated are you (1= not at all, 10=highly motivated)
- ✓ Are you ready to make the change?
 - (1) I have not thought about it before today;
 - (2) I thought about it before;
 - (3) I have thought about it and made a plan;
 - (4) I have started making a change <6months ago;
 - (5) I made the change >6months ago and still going strong
- ✓ Have you attempted to make this change before?
 - What barriers did you face when you tried to make the change?
 - Are the barriers still there? Are there other barriers?
 - i. What do you need to overcome these barriers?

Follow-Up Campus Resources

Medical School is hard. The pressure is immense. Research suggests that student burnout and mental illness are a problem in U.S. and International Medical Schools. Many medical students and physicians are not trained in their own care and often do not see their own wellness as a primary concern. This can lead to burnout and stress, which can lead to decreased academic performance, substance abuse and bad health. On the other hand, medical students and physicians with regular wellness practices perform better academically, show more compassion and have improved psychological well-being. You can take your health and well-being in your own hands to meet the high demands of medical school and set the foundation for a long and healthy professional career.

We would like to remind you of the opportunities available at SGU if you experience academic difficulties or wellness and/or mental health concerns.

Academic Advising, Development and Support (AADS)

The goal of the Academic Advising, Development and Support Division is to ensure that each student optimizes their path through Basic Sciences by taking full advantage of the resources available at St George's University.

Academic advisors are available for all SOM students who wish to receive **academic** advice during their time in Basic Sciences. Full time academic advisors are available by appointment or during open hours (8:30am – 4:30pm Mon-Fri). To schedule an appointment please send an email (academicadvisingSOM@sgu.edu); call (444 4175 ext.3027 or 3494), or visit the academic advising office (Lower David Brown Hall).

Department of Educational Services (DES)

DES is your one-stop for improving your academic performance and adapting your learning strategies. You can schedule one-to-one individual appointments with a learning strategist, MCQ approach appointments with a learning strategist (individually and in small groups of 2-3) or strategic online learning strategy support through MyCoach Med. For more information, check out the DES web site: <https://www.sgu.edu/student-support-services/department-educational-services/>

Psychological Services Center (PSC)

PSC provides psychological counselling services through scheduled and walk-in hours from 9 am to 4:30 pmAST, in addition to 24-hour crisis coverage. The PSC also provides group services for dealing with anxiety, grief and chronic conditions, and for learning Mindfulness Skills. For more information, please see the PSC web site: <https://mycampus.sgu.edu/web/psc>

References

- 1) DASS: A global measure of perceived stress. *Journal of Health and Social Behavior*, 24, 385-396.
- 2). Buysse, D. J., Reynolds, C. F, Monk, T.H., Berman, S.R., & Kupfer, D.J (1989). The Pittsburgh Sleep Quality Index (PSQI): A new instrument for psychiatric research and practice. *Psychiatry Research* 28, 193-213.